

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:08-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

08-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-10(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

PHYSICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

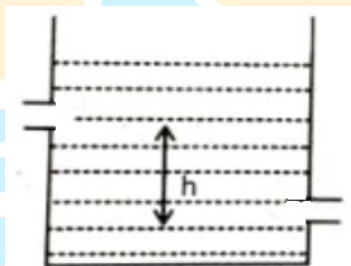
1. A wire frame in the shape of an equilateral triangle is hinged at one vertex so that it can swing freely in a vertical plane, with the plane of the triangle always remaining vertical. The side of the frame is $\frac{1}{\sqrt{3}}m$. The time period in seconds of small oscillations of the frame will be

A) $\frac{\pi}{\sqrt{2}}$ B) $\pi\sqrt{2}$ C) $\frac{\pi}{\sqrt{6}}$ D) $\frac{\pi}{\sqrt{5}}$

2. The dimensions of angular momentum are

A) $M^0L^{-1}T^0$ B) $M^0L^{-2}T^0$ C) M^1LT^{-2} D) $M^1L^2T^{-1}$

3. There are two identical small holes on the opposite sides of a tank containing liquid. The tank is open at the top. The difference in height between the two holes is h . As the liquid comes out of the two holes, the tank will experience a net horizontal force proportional to.

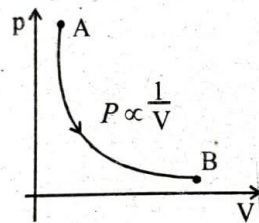


A) $\sqrt{h}$ B) h C) $h^{3/2}$ D) h^2

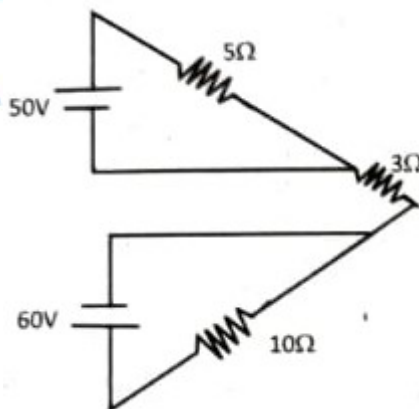
4. One end of a long metallic wire of length L is tied to the ceiling. The other end is tied to a mass less spring of force constant K . A mass m hangs freely from the free end of the spring. The area of cross-section and Young's modulus of the wire are A and Y respectively. If the mass is slightly pulled down and released, it will oscillate with a time period T equal to

A) $2\pi\sqrt{m/K}$ B) $2\pi\sqrt{\frac{m(YA+KL)}{YAK}}$ C) $2\pi\sqrt{\frac{mYA}{KL}}$ D) $2\pi\sqrt{\frac{mL}{YA}}$

5. Internal energy of the gas as it expands according to the graph AB which is a rectangular hyperbola

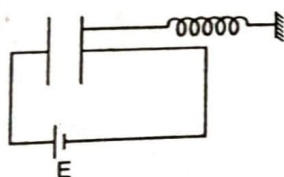


- A) Increasing continuously B) decreasing continuously
C) Always constant D) Initially Increasing then decreasing.
6. There are four concentric shells A,B,C and D of radii $a, 2a, 3a$ and $4a$ respectively. Shells B & D are given charges $+q$ & $-q$ respectively. Shell C is now earthed. The potential difference $V_A - V_C$ is ____ (take $\frac{1}{4\pi\epsilon_0} = K$)
- A) $\frac{Kq}{6a}$ B) $\frac{Kq}{2a}$ C) $\frac{Kq}{3a}$ D) $\frac{Kq}{4a}$
7. Two Identical discs initially at rest are in contact on a table. A third disc of same mass but of double radius strikes them symmetrically and itself comes to rest after impact. The co-efficient of restitution is:
- A) $\frac{9}{16}$ B) $\frac{3}{4}$ C) $\frac{1}{2}$ D) $\frac{1}{16}$
8. Find out the value of current through 3Ω resistance for the given circuit

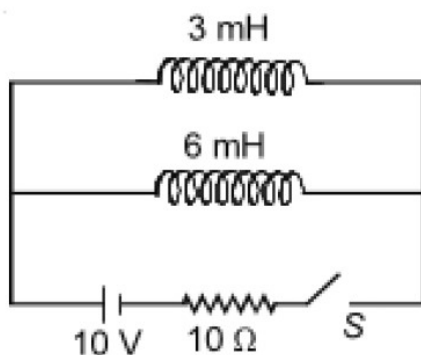


- A) 10 amp B) 6 amp C) 4 amp D) zero

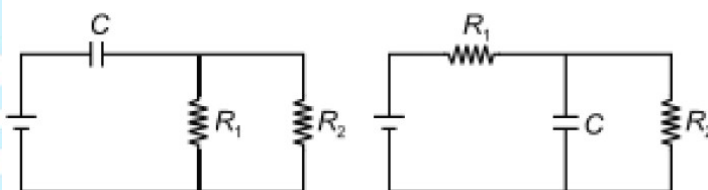
9. One plate of a capacitor is connected to a spring as shown in figure. Area of both the plates is A . In steady state separation between the plates is $0.8d$ (spring was unstretched and the distance between the plates was d when the capacitor was uncharged). The force constant of the spring is approximately



- A) $\frac{4\varepsilon_0 AE^2}{d^3}$ B) $\frac{2\varepsilon_0 AE}{d^2}$ C) $\frac{6\varepsilon_0 E}{Ad^3}$ D) $\frac{\varepsilon_0 AE^3}{2d^3}$
10. The current through 3 mH inductor in steady state after closing switch S is



- A) $\frac{1}{3}$ ampere B) $\frac{2}{3}$ ampere C) 1 ampere D) $\frac{3}{2}$ ampere
11. STATEMENT-1 : Time constants of the circuits shown in the figure are same.

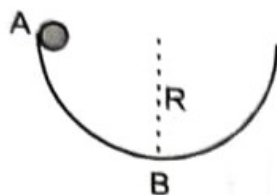


AND

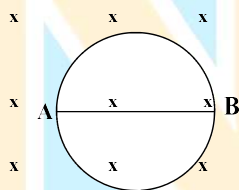
STATEMENT-2 : Instantaneous current through the capacitor branch is same at any instant for both the circuits, if batteries are inserted in the circuits at $t=0$.

- A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
- B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- C) Statement-1 is True, Statement-2 is False
- D) Statement-1 is False, Statement-2 is True

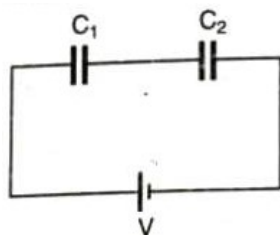
12. A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R ($\gg r$) as shown in figure. If the sphere starts from rest at the top point of the hemisphere. Find the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere.



- A) $\frac{10}{7}mg$ B) $\frac{17}{7}mg$ C) $\frac{5}{7}mg$ D) $\frac{7}{5}mg$
13. Refractive index of a prism is $\sqrt{\frac{7}{3}}$ and the angle of prism is 60° . The minimum angle of incidence of a ray that will be transmitted through the prism is _____
- A) 30° B) 40° C) 60° D) 90°
14. The radius of the conducting loop shown in figure is R . Magnetic field is decreasing at a constant rate α . Resistance per unit length of the loop is ρ . Then current in wire AB is (AB is one of the diameters)



- A) $\frac{R\alpha}{2\rho}$ from A to B B) $\frac{R\alpha}{2\rho}$ from B to A C) $\frac{2R\alpha}{\rho}$ from A to B D) zero
15. In YDSE, coherent monochromatic light having wavelength 600 nm has fallen on slits. First order bright fringe is at 4.84 mm from central maxima. Determine the wavelength for which the first order dark fringe will be observed at same location on screen? Take $D = 3m$
- A) 600 nm B) 1200 nm C) 300 nm D) 900 nm
16. Two capacitors having capacitance C_1 & C_2 are connected in series and a potential difference V is applied across them. Then:



V_1, V_2 & U_1, U_2 be the potentials drop & energy store in C_1 & C_2 respectively.

Match the entries in Column-I with Column-II :

Column-I		Column-II	
a)	$V_1 < V_2$	p)	$C_1 < C_2$
b)	$U_1 < U_2$	q)	$C_1 > C_2$
c)	V_1	r)	$C_1 V / (C_1 + C_2)$
d)	V_2	s)	$C_2 V / (C_1 + C_2)$

A) $(a \rightarrow q); (b \rightarrow q); (c \rightarrow s); (d \rightarrow r)$ B) $(a \rightarrow r); (b \rightarrow q); (c \rightarrow p); (d \rightarrow s)$

C) $(a \rightarrow q); (b \rightarrow p); (c \rightarrow r); (d \rightarrow s)$ D) $(a \rightarrow s); (b \rightarrow r); (c \rightarrow p); (d \rightarrow q)$

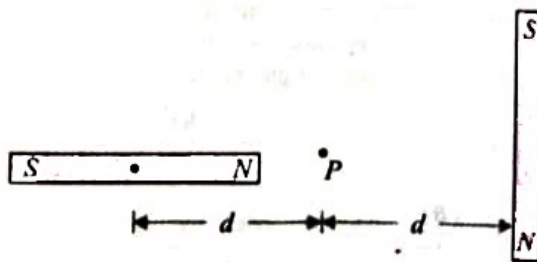
17. A changing electric field produces magnetic field. The direction of this magnetic field is

- A) In the direction of electric field
- B) In the direction opposite to the electric field
- C) Perpendicular to the direction of electric field
- D) Independent of the direction of electric field

18. A photon collides with a stationary hydrogen atom in ground state inelastically. Energy of the colliding photon is 10.2 eV. After a time interval of the order of microsecond another photon collides with same hydrogen atom inelastically with an energy of 15 eV. What will be observed by the detector?

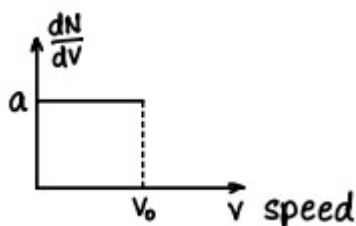
- A) 2 photons of energy 10.2 eV
- B) 2 photons of energy 1.4 eV
- C) one photon of energy 0.2 eV and an electron of energy 1.4 eV
- D) one photon of energy 10.2 eV and an electron of energy 1.4 eV

19. Two short bar magnets of magnetic moment M each are placed at a distance $2d$ apart. The magnetic field. Midway between them at P is



- A) $\frac{\mu_0}{4\pi} \frac{3M}{d^3}$
- B) $\frac{\mu_0}{4\pi} \frac{M\sqrt{5}}{d^3}$
- C) $\frac{\mu_0}{4\pi} \frac{2M}{d^3}$
- D) $\frac{\mu_0}{4\pi} \frac{M}{d^3}$

20. Graph shows a hypothetical speed distribution for a sample of N gas particles (for $V > V_0, \frac{dN}{dV} = 0$). If V_{ave} and V_{rms} are the average speed and rms speed of the gas molecules, then



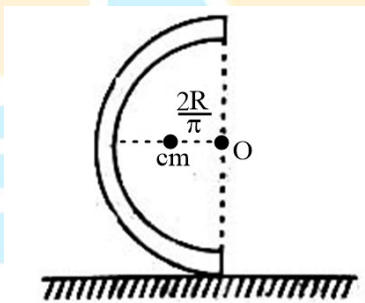
- A) $V_{ave} = \frac{V_0}{3}$ B) $V_{rms} = \frac{V_0}{\sqrt{2}}$ C) $V_{ave} : V_{rms} = 3 : \sqrt{2}$ D) $V_{ave} : V_{rms} = \sqrt{3} : 2$

SECTION-II (NUMERICAL VALUE ANSWER TYPE)

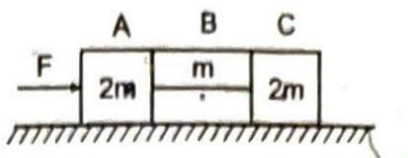
This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

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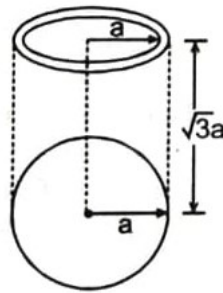
21. A half section of thin uniform pipe of mass m and radius r is released from rest. Pipe rolls without slipping. The change in PE of pipe when it has rolled through 90° is $\frac{nmgR}{\pi}$. Then the value of n is _



22. The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C friction coefficient between B and C is μ . Minimum value of F to prevent block B from downward slipping is $\left(\frac{n}{2\mu}\right)mg$. Then the value of n is _____.



23. A uniform stick of length l and mass m lies on a smooth table. It rotates with angular velocity ω about an axis perpendicular to the table and through one end of the stick. The angular momentum of the stick about the end is $\frac{ml^2\omega}{n}$. Find the value of n _____.
24. A uniform ring of mass m is lying at a distance $\sqrt{3}a$ from centre of a sphere of mass M just over the sphere where a is the radius of ring as well as that of sphere. Then, gravitational force exerted is $\frac{\sqrt{3}GMm}{na^2}$ then the value of n is _____



25. Two tuning forks A & B when sounded together produces 4 beats/s. If B is loaded with wax then also beat frequency remains same. Frequency of A is 242 Hz, find frequency of B?
26. The relative density of ice is 0.9 and that of sea water is 1.125. The fraction of the whole volume of an iceberg appears above the surface of the sea is $\frac{1}{x}$ then the value of x is _____
27. An electric current is flowing through a circular coil of radius R . The ratio of the magnetic field at the centre of the coil and that at a distance $2\sqrt{2}R$ from the centre of the coil and on its axis is _____
28. When a metallic surface is illuminated by a light of frequency $8 \times 10^{14} \text{ Hz}$, photoelectron of maximum energy 0.5 eV is emitted. When the same surface is illuminated by light of frequency $12 \times 10^{14} \text{ Hz}$, photoelectron of maximum energy 2 eV is emitted. The work function is $\frac{10}{n} \text{ (eV)}$. Then the value of n is _____
29. An unstable heavy nucleus at rest breaks in to two nuclei which moves away with velocities in the ratio of 8:27. The ratio of the radii of the nuclei (assume to be spherical) is x . Then $y = 2x$ is _____
30. In a meter bridge experiment null point is obtained at 20 cm from one end of the wire when resistance 'x' is balanced against another resistance y . If $x < y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4x$ against ' y ' in _____ cm.

CHEMISTRY

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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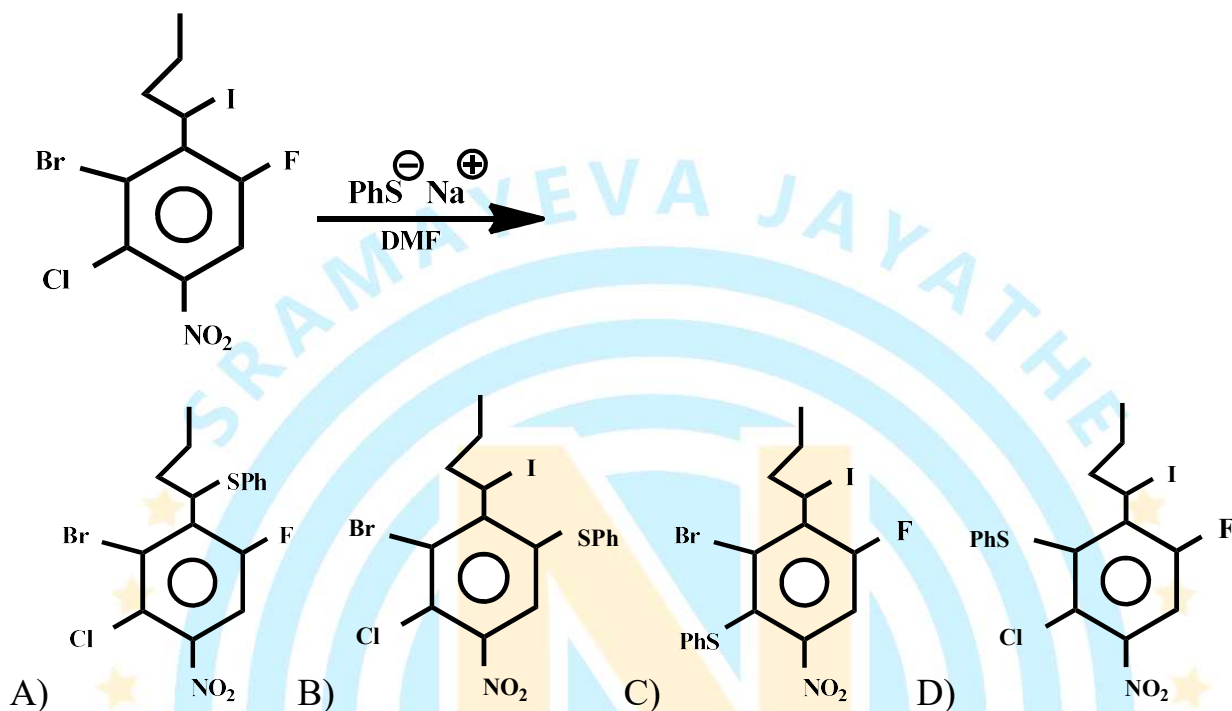
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31. Which of the following is not the name of an element with atomic number 104?
- A) Nobelium B) Rutherfordium C) Kurchatovium D) Unnilquadium
32. During halogen test, sodium fusion is boiled with *con.HNO₃* to
- A) remove unreacted sodium
B) decompose cyanide or sulphide of sodium
C) extract halogen from organic compound
D) Maintain the p^H of extract
33. In the ionization of N_2 to N_2^+ , the electron is lost from a
- A) σ -orbital B) π -orbital C) σ^* -orbital D) π^* -orbital
34. A certain substance 'A' tetramerises in water to the extent of 80%. A solution of 2.5g of 'A' in 100g of water lowers the freezing point by $0.3^\circ C$. The m.wt of 'A' is ($K_f = 1.86 \text{ k.kg.mol}^{-1}$)
- A) 122 B) 31 C) 344 D) 62
35. Which technique among the following is most appropriate in separation of a mixture of 100 mg of p-nitrophenol and picric acid?
- A) Steam distillation B) Distillation under reduced pressure.
C) Sublimation D) Thin layer chromatography.
36. The equilibrium constants K_{p_1} and K_{p_2} for the reactions $X \rightleftharpoons 2Y$ and $Z \rightleftharpoons P + Q$ respectively are in the ratio of 1:9. If the degree of dissociation of X and Z be equal then the ratio of total pressure at these equilibria is
- A) 1:36 B) 1:1 C) 1:3 D) 1:9

37. Transition metal complex with the highest value of crystal field splitting energy (Δ_0) will be

- A) $[Cr(H_2O_6)]^{+3}$ B) $[Mo(H_2O_6)]^{+3}$ C) $[Fe(H_2O_6)]^{+3}$ D) $[Os(H_2O_6)]^{+3}$

38. The major product of the reaction is



39. When an aldehyde is heated with Fehlings solution, a reddish brown precipitate is formed which is

- A) CuO B) Cu C) Cu₂O D) Cu - C $\equiv$ C - Cu

40. Which of the following cannot be prepared by using willamson's synthesis?

- A) Methoxy benzene ether B) Benzyl p-nitro phenyl ether
C) t-butyl methyl ether D) Ditertiarybutyl ether

41. Among the following the metal with the highest melting point will be

- A) Hg B) Ag C) Ga D) Cs

42. The element of group 15 which can form a strong bond with hydrogen is

- A) Nitrogen B) Phosphorous C) Arsenic D) Antimony

43. Which of the following is not a mineral of fluorine?

- A) Fluorspar B) Cryolite C) Fluoroapetite D) Carnallite

44. Consider the following statements:

According to Werner's theory

i) Secondary valencies are non-ionisable and directional

ii) Secondary valencies are satisfied by neutral molecules or negative ions .

iii) Ligands form coordinate bonds with metal ions or atoms.

iv) The charge on the complex ion is always equal to the oxidation state of the metal atom.

A) all are correct

B) (i), (ii) and (iii) are correct

C) (i), (ii) and (iv) are correct

D) (iii) and (iv) are correct

45. Zinc and mercury do not show variable valency like other d-block elements because

A) They are soft

B) Their (n-1) d-shells are completely filled

C) They have only two electrons in the outermost shell

D) Their d-shells are incompletely filled.

46. Assertion (A): The purple colour of KMnO_4 is due to charge transfer transition

Reason (R): The intense colour, in most of the transition metal complexes is due to d – d transition.

A) Both A and R are true and R is the correct explanation of A

B) Both A and R are true but R is not the correct explanation of A

C) A is true but R is false

D) A is false but R is true

47. Which of the following doesn't undergo Friedal-craft's reaction?

A) Xylene

B) Nitrobenzene

C) Cumene

D) Toluene

48. Match the transition element ions given in Column I with the characteristic (s) of products given in Column II.

Column –I

- a) Cu^{2+}
 b) Zn^{2+}
 c) Cr^{3+}
 d) Ni^{2+}

Column – II

- P) Form amphoteric oxide
 Q) Diamagnetic and colourless compounds
 R) Coloured hydrated transition metal ion
 S) Paramagnetic

- A) a – RS, b – PQ, c – PRS, d – QRS
 B) a – PQ, b – RS, c – PRS, d – QRS
 C) a – PRS, b – RS, c – PQ, d – QRS
 D) a – RS, b – PQ, c – QRS, d – PRS
49. Oxidation state of iodine in H_4IO_6^- is
 A) +7 B) +5 C) +1 D) -1
50. Which of the following does not give a white precipitate when dilute hydrochloric acid is added?
 A) Ag^+ B) Ba^{+2} C) Pb^{+2} D) Hg_2^{+2}

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

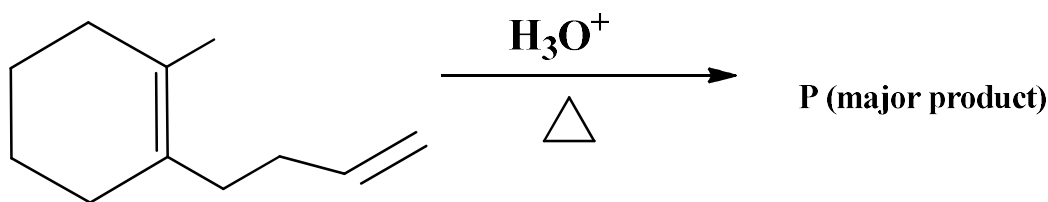
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51. The wave length of an electron and a neutron will become equal when the velocity of electron is x times the velocity of neutron. The value of x is (nearest integer)

(Mass of electron = $9.1 \times 10^{-31} \text{ kg}$), (Mass of neutron = $1.6 \times 10^{-27} \text{ kg}$)

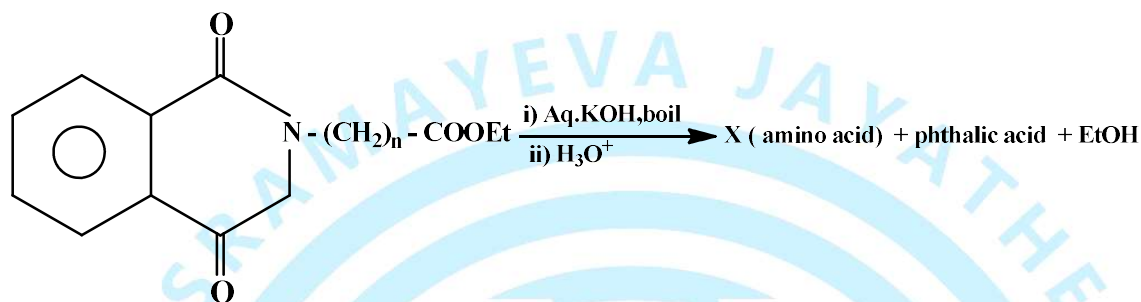
52. The major product in the following reaction is



The degree of unsaturation of product P is _____

53. The volume of $0.15M H_2SO_4$ solution required to neutralise 15 ml of $0.5M KOH$ in presence of phenolphthalein indicator is ____ ml.
54. An known amine A with benzene sulphonyl chloride yields a derivative which dissolves in aq.KOH solution. Compound A on reaction with nitrous acid gives an alcohol B which responds to iodoform test. If hydrochloride of A contains 37.2% of chlorine by weight, the total number of carbon atoms present in compounds A and B is ____.

55.



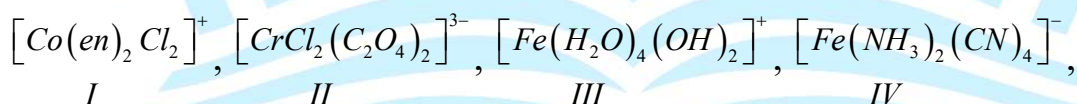
If the molecular weight of tripeptide formed by X is 189, the value of n is _____

56. Amongst the following, the number of species with only two lone pairs of electrons on central atom is ____.

SF_6 , XeF_4 , CF_4 and H_2O

57. 50 g of ethylene glycol is dissolved in 170.3 g of water and is cooled to $-9.3^\circ C$. The amount of water in g, separated as ice ($K_f = 1.86 K / \text{molal}$)

58. How many of the following can exhibit geometrical isomerism?



59. 10 ml of a gaseous organic compound (vapour density = 23) containing C,H and O only was mixed with 100 ml of oxygen and exploded under conditions which allowed the water formed to condense. The volume of gas after explosion was 90 ml. On treatment with potash solution, a further contraction of 20 ml in volume was observed. The number of moles of CO_2 formed by 1 mole of organic compound is ____

60. How many moles of $(CH_3CO)_2O$ are required to react with 1 mole of sucrose?

MATHEMATICS**MAX.MARKS: 100****SECTION – I
(SINGLE CORRECT ANSWER TYPE)**

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61. If a denotes the number of permutation of n different things taken all at a time, b the number of permutation of $n-2$ different things taken 10 at a time and c , the number of permutations of $n-12$ different things taken all at a time such that $a = 182bc$, then value of n is:
- A) 10 B) 12 C) 14 D) 18
62. STATEMENT – 1: $f(x) = |x[x]|$ is discontinuous at all Integers, where $[.]$ denotes G.I.F
STATEMENT – 2: If a function is non-differentiable at a point then it may be continuous at that point
- A) Statement 1 is true, statement 2 is true
B) Statement 1 is false, statement 2 is false
C) Statement 1 is true, statement 2 is false
D) Statement 1 is false, statement 2 is true
63. If a, b, c are real, then $f(x) = \begin{vmatrix} x+a^2 & ab & ac \\ ab & x+b^2 & bc \\ ac & bc & x+c^2 \end{vmatrix}$ is decreasing in:
- A) $\left(-\frac{2}{3}(a^2+b^2+c^2), 0\right)$ B) $\left(0, \frac{2}{3}(a^2+b^2+c^2)\right)$ C) $\left(0, \frac{(a^2+b^2+c^2)}{3}\right)$ D) Never decreases
64. If $p^{\text{th}}, q^{\text{th}}, r^{\text{th}}$, terms of a G.P are the positive numbers a, b, c respectively then angle between the vectors $\log a^3 \hat{i} + \log b^3 \hat{j} + \log c^3 \hat{k}$ and $(q-r)\hat{i} + (r-p)\hat{j} + (p-q)\hat{k}$ is:
- A) $\frac{\pi}{2}$ B) $\frac{\pi}{3}$ C) 0 D) $\sin^{-1}\left(\frac{1}{\sqrt{p^2+q^2+r^2}}\right)$

65. Consider the data on X taking the values $0, 2, 4, 8, \dots, 2^n$ with frequencies ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ respectively. If the mean of this data is $\frac{728}{2^n}$, then n is equal to ____.
- A) 15 B) 8 C) 4 D) 6
66. Complete set of real values of 'a' for which the equation $x^4 - 2ax^2 + x + a^2 - a = 0$ has all its roots real
- A) $\left[\frac{3}{4}, \infty\right)$ B) $[1, \infty)$ C) $[2, \infty)$ D) $[0, \infty)$
67. The value of $\int_{-20\pi}^{20\pi} [\sin x + \cos x] dx$ is: (where $[.]$ denotes greatest integer function)
- A) 10π B) -20π C) 20π D) -10π
68. Statement-1: The number of non-negative integral solutions of $2x + y + z = 21$ is 132
Statement-2: For $n \in N, (n^2)!$ is divisible by $(n!)^n$.
- A) Statement 1 is true, statement 2 is true
B) Statement 1 is false, statement 2 is false
C) Statement 1 is true, statement 2 is false
D) Statement 1 is false, statement 2 is true
69. Of all the functions that can be defined from the set $A: \{1, 2, 3, 4\} \rightarrow B: \{5, 6, 7, 8, 9\}$ a mapping is randomly selected. The chance that the selected mapping is strictly monotonic is
- A) $\frac{1}{105}$ B) $\frac{2}{125}$ C) $\frac{4}{4096}$ D) $\frac{5}{2048}$
70. Three distinct numbers a_1, a_2, a_3 are in increasing G.P. $a_1^2 + a_2^2 + a_3^2 = 364$ and $a_1 + a_2 + a_3 = 26$, then the value of a_{10} if a_n is the n^{th} term of the given G.P is:
- A) 2.3^9 B) 3^9 C) 2.3^{10} D) 3^{12}

71. Let $x = (5\sqrt{2} + 7)^{19}$, then $x\{x\}$ ($\{x\}$ denotes the fractional part of x) is equal to:

- A) 2^{19} B) 3^{19} C) 0 D) 1

72. If $I_a = \int_0^{\pi/2} \frac{dx}{2 \cos x + \sin x + a}$, then the value of I_a for

Column – I	Column – II
A) $a = 1$	P) $\ln\left(\frac{3}{2}\right)$
B) $a = 3$	Q) $\frac{1}{2} \log 3$
C) $a = 2$	R) $\frac{2}{\sqrt{11}} \left(\tan^{-1} \frac{3}{\sqrt{11}} - \tan^{-1} \frac{1}{\sqrt{11}} \right)$
D) $a = 4$	S) $\tan^{-1}\left(\frac{1}{3}\right)$

A) A – Q ; B – P ; C – S ; D – R

B) A – Q ; B – S ; C – R ; D – P

C) A – Q ; B – S ; C – P ; D – R

D) A – Q ; B – P ; C – Q ; D – S

73. A curve passing through (2,3) and satisfying the differential equation

$$\int_0^x ty(t) dt = x^2 y(x), (x > 0) \text{ is :}$$

A) $x^2 + y^2 = 13$

B) $y^2 = \frac{9}{2}x$

C) $\frac{x^2}{8} + \frac{y^2}{18} = 1$

D) $xy = 6$

74. Let $\frac{\tan\left(\frac{\pi}{4} + \alpha\right)}{5} = \frac{\tan\left(\frac{\pi}{4} + \beta\right)}{3} = \frac{\tan\left(\frac{\pi}{4} + \gamma\right)}{2}$. Then $12 \sin^2(\alpha - \beta) + 15 \sin^2(\beta - \gamma) - 7 \sin^2(\gamma - \alpha)$

is equal to

A) $-\frac{1}{2}$

B) $\frac{1}{2}$

C) 1

D) 0

75. A 13 ft. ladder is leaning against a wall when its base starts to slide away. At the instant when the base is 12 ft. away from the wall, the base is moving away from the wall at the rate of 5 ft/sec. The rate at which the angle θ between the ladder and the ground is changing is
- A) $-\frac{12}{13} \text{ rad/sec}$ B) -1 rad/sec C) $-\frac{13}{12} \text{ rad/sec}$ D) $-\frac{10}{13} \text{ rad/sec}$
76. Two aero planes I and II bomb a target in succession. The probabilities of I and II scoring a hit correctly are 0.3 and 0.2, respectively. The second plane will bomb only if the first misses the target. The probability that the target is hit by the second plane is
- A) 0.06 B) 0.14 C) 0.312 D) 0.70
77. The area of the region bounded by the curve $y = \frac{16-x^2}{4}$ and $y = \sec^{-1}[-\sin^2 x]$ (where $[.]$ denotes greatest integer function is:)
- A) $\frac{1}{3}(4-\pi)^{3/2}$ B) $8(4-\pi)^{3/2}$ C) $\frac{8}{3}(4-\pi)^{3/2}$ D) $\frac{8}{3}(4-\pi)^{1/2}$
78. If $f(x) = \frac{e^{[x]+|x|} - 3}{[x] + |x| + 1}$, then: (where $[.]$ represents greatest integer function)
- A) $\lim_{x \rightarrow 0^+} f(x) = -2$ B) $\lim_{x \rightarrow 0^-} f(x) = 0$ C) $\lim_{x \rightarrow 0^-} f(x) = 2$ D) $\lim_{x \rightarrow 0} f(x)$ exist
79. Let $g(x) = \frac{1}{4}f(2x^2 - 1) + \frac{1}{2}f(1 - x^2) \forall x \in R$, where $f''(x) > 0 \forall x \in R$, $g(x)$ is necessarily increasing in the interval
- A) $\left(-\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}\right)$ B) $\left(-\sqrt{\frac{2}{3}}, 0\right) \cup \left(\sqrt{\frac{2}{3}}, \infty\right)$
- C) $(-1, 1)$ D) None of these
80. The largest integral value of x satisfying the inequality $(\tan^{-1}(x))^2 - 4(\tan^{-1}(x)) + 3 > 0$ is:
- A) 0 B) 1 C) 2 D) 3

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Let O be an interior point of $\triangle ABC$ such that $\vec{OA} + 2\vec{OB} + 3\vec{OC} = \vec{0}$ then ratio of the area of $\triangle ABC$ to the area of $\triangle AOC$ is _____.
82. Find number of integral values of k for which the line $3x + 4y - k = 0$, lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 12y + 113 = 0$, without cutting a chord on either of circle.
83. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 4\hat{k}$, $\vec{b} = 2(\hat{i} + \hat{k})$ and $\vec{c} = 4\hat{i} + 2\hat{j} + 3\hat{k}$. If the equation $x\vec{a} + y\vec{b} + z\vec{c} = \alpha(\hat{x}\hat{i} + \hat{y}\hat{j} + \hat{z}\hat{k})$ has a non-trivial solution, then find the sum of all distinct possible values of α .
84. For each positive integer n, consider the point P with abscissa n on the curve $y^2 - x^2 = 1$. If d_n represents the shortest distance from the point P to the line $y = x$ then $\lim_{n \rightarrow \infty} (n.d_n)$ has value $\frac{1}{K\sqrt{K}}$, then K is _____.
85. Let $f(x) = \int_0^x 3^t (3^t - 4)(x - t) dt$ ($x \geq 0$). If $x = a$ is the point where $f(x)$ attains its local minimum value then value of 3^a is _____.
86. The distance between $x - \frac{y}{2} + \frac{3}{2} = 0$ & $2x - y + 4 = 0$ is α , and distance between $x - \frac{y}{2} + \frac{3}{2} = 0$ & $3x - \frac{3y}{2} + \frac{15}{2} = 0$ is β , then find $\alpha^2 + \beta^2$.
87. The number of points of intersection of $|Z - (4 + 3i)| = 2$ and $|Z| + |Z - 4| = 6, Z \in \mathbb{C}$ is _____.
88. Defined a "good word" as a sequence of letters that consists only of the letters A, B and C and in which A never immediately followed by B, B is never immediately followed by C, and C is never immediately followed by A. If the number of n-letter "good words" is 384, find n.
89. If $2ab^2 - 4ab = 6b^2, a > 0, b > 0$ then find the minimum value of $\frac{ab}{6}$.
90. If $\int \frac{x^2 + 4}{x^4 + 16} dx = \frac{1}{a\sqrt{b}} \tan^{-1} \left(\frac{x^2 - 4}{ax\sqrt{b}} \right) + c$ then $a + b$ is _____.